

Unit 1: Introduction to Machine Learning

Course: Machine Learning

Unit: 1 – Introduction to Machine Learning

Course Outcome : Understand the fundamental concepts of Machine Learning.

Learning Objectives

After completing this unit, students should be able to:

- Understand the concept of Machine Learning (ML).
- Differentiate between AI, ML, and Deep Learning.
- Explain different learning types.
- Describe supervised learning with suitable examples.
- Understand the concepts of VC Dimension and PAC Learning.
- Explain the impact of noise in learning.
- Differentiate binary and multiclass classification.
- Explain regression techniques.
- Understand model selection and generalization.
- Identify the dimensions/components of a supervised ML algorithm.

1. Introduction to Machine Learning

Definition

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data without being explicitly programmed.

Arthur Samuel (1959):

"Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed."

Tom Mitchell (1997):

"A computer program learns from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T improves with experience E."

Relationship between AI, ML and Deep Learning

Artificial Intelligence

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|----- Machine Learning

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|----- Deep Learning

- AI → Broad concept of intelligent machines
- ML → Subset of AI that learns from data
- DL → Subset of ML using neural networks

Traditional Programming vs Machine Learning

Traditional Programming

Input + Program → Output

Example:

Marks + Grade Rules → Grade

Machine Learning

Input + Output → Learning Algorithm → Model

Model + New Input → Prediction

Why Machine Learning?

Traditional programming becomes difficult when:

- Rules are too complex
- Huge amount of data exists
- Patterns keep changing
- Manual programming is impossible

ML automatically discovers hidden patterns from data.

Characteristics of Machine Learning

- Learns from experience
- Improves performance over time
- Finds hidden patterns
- Makes predictions
- Handles large datasets

- Works with uncertain environments

2. Examples of Machine Learning Applications

Healthcare

- Disease prediction
- Medical image analysis
- Drug discovery

Example:

Predict diabetes using patient health records.

Banking

- Fraud detection
- Credit score prediction
- Loan approval

E-commerce

- Product recommendation
- Customer segmentation
- Price prediction

Example:

Amazon recommends products.

Entertainment

- Movie recommendation

Example:

Netflix suggests movies.

Email Services

- Spam detection
- Smart reply

Agriculture

- Crop prediction
- Disease detection

Transportation

- Self-driving cars
- Traffic prediction

Social Media

- Face recognition
- Friend recommendation

3. Types of Machine Learning

Machine Learning

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|---- Supervised Learning

|

|---- Unsupervised Learning

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|---- Semi-supervised Learning

|

|---- Reinforcement Learning

A. Supervised Learning

Training data contains:

Input + Correct Output (Label)

Goal:

Learn mapping

Input → Output

Examples:

- Email → Spam/Not Spam

- Image → Cat/Dog
- Student Marks → Grade

Algorithms

- Decision Tree
- SVM
- Logistic Regression
- Random Forest
- Neural Networks

B. Unsupervised Learning

Training data has:

Only Inputs

Goal:

Discover hidden patterns

Applications

- Clustering
- Market Basket Analysis
- Customer Segmentation

Algorithms

- K-Means
- Hierarchical Clustering
- DBSCAN

C. Semi-supervised Learning

Uses

- Small labelled data
- Large unlabelled data

Example

100 labelled images

10,000 unlabelled images

D. Reinforcement Learning

Learning by trial and error.

Agent receives:

- Reward
- Penalty

Applications

- Robotics
- Game playing
- Autonomous vehicles

Comparison

Learning Type	Labelled Data	Goal
Supervised	Yes	Prediction
Unsupervised	No	Pattern Discovery
Semi-supervised	Partial	Better Accuracy
Reinforcement	Reward	Decision Making

4. Supervised Learning – Learning a Class from Examples

A classifier learns from labelled examples.

Example Dataset

Age	Income	Buy Laptop
-----	--------	------------

20	25K	No
----	-----	----

25	40K	Yes
----	-----	-----

35	70K	Yes
----	-----	-----

22	30K	No
----	-----	----

The algorithm learns a decision boundary.

New Customer

Age = 28

Income = 45K

Prediction → Buy Laptop = Yes

Classification Process

Training Data

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Feature Extraction

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Learning Algorithm

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Classifier Model

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Prediction

Terminology

Feature

Input variable

Example

Height

Weight

Age

Label

Output variable

Example

Disease

Spam

Fraud

Training Set

Used to train the model.

Testing Set

Used to evaluate the model.

5. Vapnik-Chervonenkis (VC) Dimension

Definition

VC Dimension measures the learning capacity (complexity) of a model.

Higher VC Dimension

↓

More complex model

↓

Can classify more patterns

Example

A straight line

VC Dimension = 3

Meaning

Any arrangement of 3 points can be classified.

But not every arrangement of 4 points.

Importance

- Measures model complexity
- Helps prevent overfitting
- Used in statistical learning theory

6. Probably Approximately Correct (PAC) Learning

PAC Learning provides a theoretical framework to determine whether a learning algorithm can learn effectively from examples.

Two Important Terms

Probably

Confidence

Usually

95%

99%

Approximately Correct

Prediction need not be perfect.

Small acceptable error is allowed.

PAC Learning Conditions

Algorithm should

- Learn from finite examples
- Produce low error
- Have high confidence

Importance

- Measures learning efficiency
- Estimates sample complexity
- Guides algorithm design

7. Noise in Machine Learning

Noise refers to unwanted or incorrect information in data.

Types of Noise

Attribute Noise

Incorrect feature values

Example

Age = 250 years

Class Noise

Wrong labels

Dog image labelled as Cat

Missing Values

Salary = NULL

Random Noise

Sensor errors

Typing mistakes

Effects of Noise

- Lower accuracy
- Wrong predictions
- Overfitting
- Poor generalization

Handling Noise

- Data cleaning
- Remove duplicates
- Outlier detection
- Feature selection
- Robust algorithms

8. Learning Multiple Classes

Binary Classification

Only two classes

Example

Spam / Not Spam

Multi-class Classification

More than two classes

Example

Animal Classification

Cat

Dog

Horse

Elephant

Approaches

One-vs-All (OvA)

One classifier for each class versus all others.

One-vs-One (OvO)

One classifier for every pair of classes.

9. Regression

Regression predicts continuous numerical values.

Examples

- House price prediction
- Temperature prediction
- Salary prediction

Difference between Classification and Regression

Classification	Regression
Predict Category	Predict Value
Output Label	Output Number
Spam Detection	House Price

Common Algorithms

- Linear Regression
- Polynomial Regression
- Decision Tree Regression

10. Model Selection and Generalization

Model Selection

Choosing the best model among several alternatives based on performance.

Criteria:

- Accuracy
- Precision
- Recall
- F1-score
- Computational cost

Generalization

Ability of a model to perform well on unseen data.

Good Model

Training Accuracy $\approx$ Testing Accuracy

Overfitting

Training Accuracy = 100%

Testing Accuracy = 70%

Model memorizes data.

Poor generalization.

Underfitting

Training Accuracy = 60%

Testing Accuracy = 58%

Model is too simple.

Good Fit

Training Accuracy = 92%

Testing Accuracy = 90%

Excellent generalization.

Techniques to Improve Generalization

- Cross-validation
- Regularization (L1/L2)
- Early stopping
- Feature selection
- More training data
- Hyperparameter tuning

11. Dimensions of a Supervised Machine Learning Algorithm

A supervised ML system consists of the following dimensions/components:

1. Training Data

- Labelled examples used to learn patterns.

2. Features

- Input variables that describe the data.

3. Target Variable

- Desired output or class label.

4. Hypothesis Space

- Set of all possible models the algorithm can choose from.

5. Learning Algorithm

- Procedure that finds the best hypothesis.

6. Loss Function

- Measures prediction error (e.g., Mean Squared Error, Cross-Entropy).

7. Optimization Algorithm

- Minimizes the loss function (e.g., Gradient Descent).

8. Evaluation Metric

- Measures performance (Accuracy, Precision, Recall, RMSE).

9. Generalization Ability

- Performance on unseen data.

Summary of Unit

- Machine Learning enables computers to learn from data.
- ML is widely used in healthcare, banking, transportation, and e-commerce.
- Four main learning paradigms are supervised, unsupervised, semi-supervised, and reinforcement learning.
- Supervised learning uses labelled data to build predictive models.
- VC Dimension measures model capacity, while PAC Learning provides a theoretical framework for learnability.
- Noise can degrade model performance and should be handled through data preprocessing.
- Classification predicts discrete labels, whereas regression predicts continuous values.
- Model selection and generalization are essential for building accurate and robust models.
- A supervised ML algorithm comprises training data, features, target variables, hypothesis space, learning algorithms, loss functions, optimization methods, evaluation metrics, and generalization capability.

Classroom Discussion Questions

1. How is Machine Learning different from traditional programming?
2. Give three real-life applications of Machine Learning.
3. Compare supervised and unsupervised learning with examples.
4. What is the significance of VC Dimension?
5. Explain PAC Learning in simple terms.
6. What are different types of noise in data?
7. Distinguish between binary and multiclass classification.
8. Differentiate regression and classification with suitable examples.
9. What are overfitting and underfitting? How can they be reduced?
10. List and explain the major components of a supervised Machine Learning algorithm.

Suggested Activities

- **Activity 1:** Identify five Machine Learning applications used in everyday life and classify them by learning type.
- **Activity 2:** Use a small labelled dataset (e.g., student marks and grades) to demonstrate supervised learning.
- **Activity 3:** Compare classification and regression using simple datasets and discuss the expected outputs.

- **Activity 4:** Discuss examples of noisy data and suggest preprocessing techniques to improve data quality.
- **Activity 5:** Analyze examples of overfitting and underfitting by comparing training and testing performance on simple models.

A complete set of theory notes for the entire Unit 1 is too long to fit into a single response. I'll begin with **Part 1: Introduction to Machine Learning** in detailed lecture-note format. The remaining topics can be prepared similarly in subsequent parts.

UNIT 1: INTRODUCTION TO MACHINE LEARNING

Detailed Theory Notes (Part 1)

1. Introduction to Machine Learning

1.1 What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed for every task. Instead of following fixed rules, ML algorithms identify patterns in historical data and use these patterns to make predictions or decisions on new data.

Machine Learning has become an essential technology because modern organizations generate enormous amounts of data that cannot be effectively analyzed using traditional programming techniques. ML systems automatically discover relationships within the data, making them suitable for applications such as fraud detection, medical diagnosis, recommendation systems, and autonomous vehicles.

Definitions

Arthur Samuel (1959):

"Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed."

Tom M. Mitchell (1997):

"A computer program is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P** if its performance at tasks in **T**, as measured by **P**, improves with experience **E**."

Explanation of Mitchell's Definition

- **Task (T):** The activity to be performed (e.g., classifying emails as spam or not spam).
- **Experience (E):** The training data used by the algorithm.
- **Performance Measure (P):** A metric used to evaluate performance (e.g., accuracy, precision, recall).

If the algorithm performs better after learning from the data, it is considered to have learned.

1.2 Evolution of Machine Learning

The development of Machine Learning has evolved over several decades.

Year	Milestone
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1950	Alan Turing proposed the Turing Test
1952	Arthur Samuel developed a self-learning checkers program
1957	Frank Rosenblatt introduced the Perceptron
1986	Backpropagation algorithm popularized neural networks
1997	IBM Deep Blue defeated world chess champion Garry Kasparov
2012	Deep Learning revolution began with AlexNet
Present	Widespread use of AI in healthcare, finance, transportation, and education

1.3 Artificial Intelligence, Machine Learning, and Deep Learning

Artificial Intelligence, Machine Learning, and Deep Learning are closely related but differ in scope.

Artificial Intelligence (AI)

AI is the broad field of creating systems that perform tasks requiring human intelligence, such as reasoning, problem-solving, learning, and decision-making.

Examples:

- Expert systems
- Robotics
- Natural language processing

Machine Learning (ML)

Machine Learning is a subset of AI where systems learn patterns from data instead of relying solely on predefined rules.

Examples:

- Spam filtering
- House price prediction
- Credit risk assessment

Deep Learning (DL)

Deep Learning is a subset of ML that uses multi-layered artificial neural networks to learn complex patterns from large datasets.

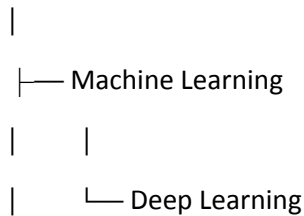
Examples:

- Facial recognition
- Speech recognition

- Autonomous driving

Relationship

Artificial Intelligence



1.4 Traditional Programming vs Machine Learning

Traditional Programming

In conventional programming, the programmer explicitly writes rules to solve a problem.

Input Data + Program → Output

Example:

A grading program where rules such as "Marks $\geq$ 90 = Grade A" are manually coded.

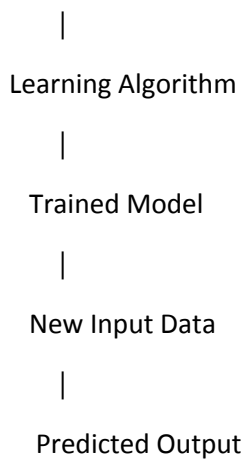
Limitations

- Difficult to handle complex problems.
- Rules become impossible to define when data is highly variable.
- Requires continuous manual updates.

Machine Learning Approach

Instead of manually writing rules, the algorithm learns them from data.

Input Data + Expected Output



Example:

Instead of programming all spam-detection rules, the model learns them from thousands of labeled emails.

Advantages over Traditional Programming

- Learns automatically from data.
- Improves with experience.
- Adapts to changing environments.
- Handles large and complex datasets.

1.5 Why Machine Learning?

Machine Learning is preferred when:

1. The amount of data is very large.
2. Writing explicit rules is difficult or impossible.
3. Patterns change over time.
4. High prediction accuracy is required.
5. Automation is needed for repetitive tasks.

Real-Life Examples

- Email spam detection
- Online shopping recommendations
- Medical diagnosis
- Fraud detection in banking
- Traffic prediction
- Voice assistants

1.6 Characteristics of Machine Learning

A Machine Learning system generally possesses the following characteristics:

1. Learns from Data

The system improves its knowledge using historical data.

2. Improves Performance

Performance increases as more training data becomes available.

3. Pattern Recognition

ML algorithms identify hidden relationships and patterns in data.

4. Prediction Capability

Once trained, the model predicts outcomes for unseen data.

5. Adaptability

ML models can be retrained to adapt to changing environments.

6. Automation

Reduces the need for manual intervention.

1.7 Machine Learning Workflow

A typical Machine Learning project follows these steps:

Problem Definition

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Data Collection

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Data Preprocessing

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Feature Engineering

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Model Selection

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Model Training

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Model Evaluation

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Prediction/Deployment

Step 1: Problem Definition

Clearly define the objective.

Example:

Predict whether a customer will purchase a product.

Step 2: Data Collection

Gather relevant historical data.

Sources:

- Databases
- Sensors
- Web applications
- CSV files
- APIs

Step 3: Data Preprocessing

Prepare the data by:

- Removing duplicates
- Handling missing values
- Correcting inconsistencies
- Normalizing features

Step 4: Feature Engineering

Select or create meaningful input variables.

Example:

From a date field, derive "Day of Week" or "Month."

Step 5: Model Selection

Choose an appropriate algorithm.

Examples:

- Decision Tree
- Logistic Regression
- Support Vector Machine
- Random Forest

Step 6: Model Training

Train the model using labeled data.

Step 7: Model Evaluation

Assess performance using metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

Step 8: Deployment

Deploy the trained model for real-world use.

1.8 Advantages of Machine Learning

- Automates decision-making processes.
- Handles large datasets efficiently.
- Identifies hidden patterns.
- Improves prediction accuracy.
- Supports real-time analytics.
- Enables personalized recommendations.
- Reduces manual effort.

1.9 Limitations of Machine Learning

- Requires high-quality data.
- Training can be computationally expensive.
- Complex models may be difficult to interpret.
- Can overfit if not properly validated.
- Sensitive to biased or noisy data.
- Requires periodic retraining as data evolves.

1.10 Real-World Applications

Healthcare

- Disease diagnosis
- Medical image analysis
- Drug discovery

Banking

- Fraud detection
- Credit scoring
- Loan approval

Retail and E-commerce

- Product recommendation
- Customer segmentation

- Demand forecasting

Transportation

- Autonomous vehicles
- Route optimization
- Traffic prediction

Education

- Personalized learning
- Student performance prediction
- Intelligent tutoring systems

Manufacturing

- Predictive maintenance
- Quality inspection
- Production optimization

Key Terms

- **Artificial Intelligence (AI):** Broad field of creating intelligent systems.
- **Machine Learning (ML):** AI technique where systems learn from data.
- **Deep Learning (DL):** ML approach using deep neural networks.
- **Model:** The learned representation used for prediction.
- **Training Data:** Data used to teach the model.
- **Feature:** Input variable describing an instance.
- **Label:** Expected output or target value.
- **Prediction:** Output produced by the trained model.

Summary

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data rather than relying solely on explicitly programmed rules. It plays a crucial role in solving complex problems across healthcare, finance, retail, transportation, and many other domains. Understanding the foundational concepts, workflow, characteristics, advantages, and limitations of Machine Learning provides the basis for studying advanced topics such as supervised learning, regression, model evaluation, and statistical learning theory.

This is **Part 1 (approximately 8–10 lecture pages)**. The next sections can cover:

1. **Examples of Machine Learning Applications**
2. **Types of Learning (Supervised, Unsupervised, Semi-supervised, Reinforcement)**
3. **Learning a Class from Examples**
4. **VC Dimension**
5. **PAC Learning**
6. **Noise in Machine Learning**
7. **Learning Multiple Classes**
8. **Regression**
9. **Model Selection and Generalization**
10. **Dimensions of a Supervised Machine Learning Algorithm**